The Student-t Multi-Fidelity Objective

`zrg_cost_function_student_t`

The full, explicit scalar function cost_fn(x) that L-BFGS-B minimizes over the 19 free parameters x in the MF Student-t runs (e.g. aurora_mf_ne128_ne32_prod_studentt_1000starts_annual.yaml), built up step by step, with this run's actual configuration (dof=4, kappa=0, lambda=0, and the PCP/OSR/OLR/TLWP weights) plugged in throughout.

Step 1 — the per-location Student-t NLL term

For predicted mean p, observed value o, and GP predictive variance v at one (location, variable) pair, with residual r = p − o:

$$\ell(r, v) = \tfrac{1}{2}\log v_c \;+\; \tfrac{\nu+1}{2}\log\!\left(1 + \tfrac{r^2}{\nu v_c}\right) \;+\; \mathrm{const}(\nu) \;+\; \kappa v_c$$

where $v_c = \max(v,\, 10^{-6})$, and

$$\mathrm{const}(\nu) = \;-\ln\Gamma\!\left(\tfrac{\nu+1}{2}\right) + \ln\Gamma\!\left(\tfrac{\nu}{2}\right) + \tfrac{1}{2}\ln(\nu\pi)$$

is the Student-t normalizing constant — fixed once ν is chosen, independent of r or v. It does not move the argmin, but makes the returned value a true negative log-likelihood.

This run's numbers: $\nu = 4,\ \kappa = 0$

The κ·v_c term vanishes. const(4) = −lgamma(2.5) + lgamma(2) + 0.5·log(4π) ≈ −0.2847 + 0 + 1.2655 ≈ 0.9808, so:

$$\ell(r, v) = \tfrac{1}{2}\log v_c + 2.5\log\!\left(1 + \tfrac{r^2}{4 v_c}\right) + 0.9808$$

As $\nu \to \infty$ this converges exactly to the Gaussian NLL term $\tfrac{1}{2}\tfrac{r^2}{v_c} + \tfrac{1}{2}\log(2\pi v_c)$ — Student-t is the same Gaussian-variance idea with heavier tails, recovering Gaussian exactly in the limit.

Step 2 — mean over locations, within one block, one variable

l is computed at every (zonal band, region, global point) location. Within one ZRG block (zonal, or regional) and one variable, the per-location values are averaged — a plain mean, since zonal_weights and regional_weights are both null in this config. The global block has a single location, so no averaging is needed — it's just l evaluated once.

Zonal and regional locations are never pooled into the same mean — they are sliced apart before any averaging happens (feature layout: [zonal_0 ... zonal_14, region_0 ... region_5, global]).

Step 3 — weighted sum over the 4 variables

At each block level, the per-variable means are combined via a weighted sum:

$$\text{zonal_block} = \sum_v w_v \cdot \overline{\ell}_{\text{zonal}}(v)$$

$$\text{regional_block} = \sum_v w_v \cdot \overline{\ell}_{\text{regional}}(v)$$

$$\text{global_block} = \sum_v w_v \cdot \ell_{\text{global}}(v)$$

This run's variable weights:

PCP	0.3	
OSR	0.1	(downweighted)
OLR	0.3	
TLWP	0.3	

Step 4 — weighted sum over zonal / regional / global

$$\text{snap_cost} = \tfrac{1}{3}\,\text{zonal_block} + \tfrac{1}{3}\,\text{regional_block} + \tfrac{1}{3}\,\text{global_block}$$

A fixed 3-way split, not proportional to location count — without it the 15 zonal bands would numerically dominate the 6 regions and especially the single global point, just by outnumbering them.

Step 5 — weighted sum over snapshots

$$\text{total_cost} = \sum_{\text{snapshots}} dy_w[\text{snap}] \cdot \text{snap_cost}[\text{snap}]$$

One snapshot here (ANN, weight 1.0), so this is a no-op: total_cost = snap_cost.

Step 6 — the training-distance penalty (inert in this run)

The full cost_fn(x) adds one more term on top:

$$\text{cost_fn}(x) = \text{total_cost} + \lambda \cdot \max(0,\ d(x) - \tau)$$

d(x) is x's distance to the nearest real training point (each parameter's own bounds_from_data-rescaled space); τ is the training ensemble's own worst-case internal spacing. This run sets λ = 0, so this term is identically zero for every x — it does not affect this run's optimization, but is structurally part of every cost_fn call.

The full assembled objective

Putting Steps 1–6 together, for this run's configuration:

$$\text{cost_fn}(x) = \tfrac{1}{3}\sum_v w_v \overline{\ell}_{15\,\text{zonal}} + \tfrac{1}{3}\sum_v w_v \overline{\ell}_{6\,\text{regions}} + \tfrac{1}{3}\sum_v w_v\, \ell_{\text{global}}$$

with $\ell(r, v) = \frac{1}{2}\log v_c + 2.5\log(1 + \frac{r^2}{4v_c}) + 0.9808$,
$(m(x), v(x)) = \texttt{gp.predict}(x)$ the MF AR1 GP's HF posterior mean/variance, $r = m(x) - \text{obs}$, $v_c = \max(v(x), 10^{-6})$.

This single scalar is what all 1000 L-BFGS-B starts independently minimize over the 19 free parameters.

This run's configuration, at a glance

Likelihood	student_t
Degrees of freedom ν	4
Variance floor	1e-6
κ (variance penalty)	0 (off)
λ (distance penalty)	0 (off)
n_xstarts	1000
max_workers	192